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Research Article

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Is Stephen Curry really a guard? New perspective on players typology using functional data analysis

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Abstract: We present a novel representation of NBA players' shooting patterns based on Functional Data Analysis (FDA). Each player's charts of made and missed shots are treated as smooth functional data defined over a two-dimensional domain corresponding to the offensive half-court. This continuous representation enables a parsimonious multivariate functional principal components analysis (MFPCA) decomposition, producing a set of common principal component functions that capture the primary modes of variability in shooting patterns, along with player-specific scores that quantify individual deviations from the average behavior. We first interpret the principal component functions to characterize the main sources of variation in shooting tendencies. We then apply k-medoids clustering to the principal component scores to construct a data-driven taxonomy of players. Comparing our empirical clusters to conventional NBA position labels reveals low agreement, suggesting that our shooting-pattern representation might capture aspects of playing style not fully reflected in official designations. The proposed methodology provides a flexible, interpretable, and continuous framework for analyzing player tendencies, with potential applications in coaching, scouting, and historical player or match comparisons.

Keywords: Basketball, Clustering, Shooting patterns, Sports Analytics

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